

Zorba Health: Consent-Aware Voice Healthcare AI with FHIR-Grounded RAG and an MCP Tool Gateway

Samarpan Koirala Independent Researcher; Maintainer, Zorba Health Open Source Project
Email: samarpankoirala@gmail.com

Abstract—Voice interfaces can broaden access to digital health support, yet conversational agents that touch clinical context introduce risks of privacy leakage, missing consent, hallucinated advice, and uncontrolled tool use. We present Zorba Health, an open-source microservice platform that combines a LiveKit-backed voice agent, a Model Context Protocol (MCP) tool gateway, a FHIR-compatible retrieval-augmented generation (RAG) path with citations, and an append-only audit and consent service. We release a synthetic evaluation corpus of 16 patients (646 filtered FHIR resources), 25 gold QA items, 22 exact and 22 paraphrase escalation utterances, and 8 consent scenarios. Offline type-aware lexical Recall@3 = 1.00. Exact-phrase escalation F1 = 1.00, but paraphrase F1 = 0.29 (combined F1 = 0.71). Consent-gate e2e deny/allow/regrant for health-record access passes, with a documented gap for third-party model consent. A live OpenAI RAG evaluation with answer-string-required scoring yields answerable accuracy = 0.85 (17/20), citation presence = 1.00, unanswerable safe refusal = 1.00 (5/5), mean latency 1.67 s (p95 2.43 s). We argue that separating voice runtime, tool policy, clinical retrieval, and compliance logging is a practical path toward auditable healthcare agents.

Index Terms—Healthcare AI, voice agents, FHIR, retrieval-augmented generation, Model Context Protocol, consent, audit, microservices, Synthea

I. Introduction

Telephone and voice channels remain highly accessible for patients who struggle with app-centric portals. Large language models (LLMs) enable natural dialogue, but naïve deployments can invent clinical facts, bypass consent, omit audit trails, and couple the agent directly to sensitive stores. The Model Context Protocol (MCP) standardizes tool exposure but cannot enforce consent or authorization at the protocol layer; implementors must build those controls [1]. WHO guidance on large multi-modal models for health similarly emphasizes governance and careful escalation [2].

Problem. How can a voice healthcare assistant be engineered so that tool use, record retrieval, and AI summarization are consent-gated, auditable, grounded in patient-specific FHIR context, and reproducibly evaluable?

Contributions. (1) an open-source voice healthcare microservice architecture; (2) a controlled MCP tool gateway that materializes the consent flows the MCP specification leaves to implementors; (3) a FHIR-compatible RAG pipeline with citations; (4) a safety, consent, and

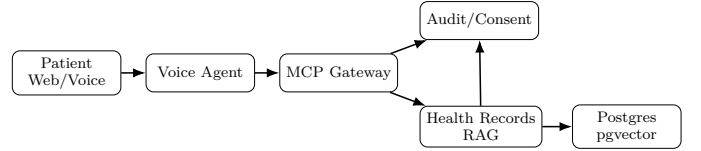


Fig. 1. High-level architecture with MCP boundary.

audit framework; (5) a reproducible synthetic evaluation package including live RAG, consent-gate, and paraphrase safety tests.

II. Related Work

Relational agents targeted accessible care [3], while modern medical LLMs encode clinical knowledge [4], [5]. Polaris [6] is a proprietary multi-agent voice constellation evaluated by over 1,100 U.S. licensed nurses; Zorba Health instead prioritizes an open-source, consent-gated MCP architecture without clinical-parity claims. HL7 FHIR R4 and Synthea [7]–[9] enable synthetic EHR research without PHI; MedAgentBench [10] adds a FHIR-compliant interactive agent benchmark. Retrieval-augmented generation [11], medical RAG benchmarks [12], [13], Almanac [14], and FHIR-RAG-MEDS [15] quantify clinical grounding. Tool-using LLMs [16], [17], AgentClinic [18], and HealthBench [19] motivate interactive evaluation beyond static QA. MCP [1] standardizes tool exposure; early healthcare MCP prototypes [20], [21] document compliance layers, while Hou et al. [22] systematize MCP threats. HIPAA [23] and WHO LMM guidance [2] motivate consent and audit. Crisis-detection benchmarks [24], [25] show LLMs can reach high F1 on explicit suicidal ideation yet still fail on indirect signals—framing our paraphrase safety collapse.

III. System Architecture

Zorba Health comprises an API gateway, voice-agent service, MCP server, domain microservices, and a dedicated audit service, with PostgreSQL (pgvector), Redis, and RabbitMQ (Fig. 1). Voice sessions do not query FHIR stores directly; tools are mediated by MCP (Fig. 2).

A. Consent-Aware RAG

The pipeline validates inputs; checks HEALTH_RECORD_ACCESS; embeds the query;

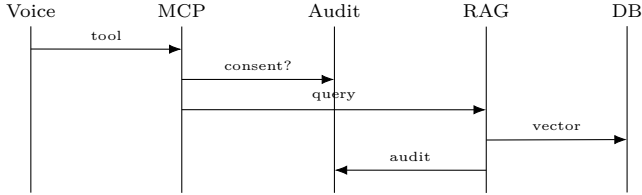


Fig. 2. Voice $\rightarrow$ MCP $\rightarrow$ consent $\rightarrow$ RAG $\rightarrow$ audit sequence.



Fig. 3. Dataset construction pipeline.

TABLE I
Evaluation corpus resource counts (filtered).

Resource type	Count
Patient	16
Condition	126
Observation	206
MedicationRequest	97
AllergyIntolerance	16
Encounter	80
DiagnosticReport	60
DocumentReference	20
CarePlan	25
Total	646

performs patient-scoped vector search; optionally summarizes only over retrieved chunks; audits events; returns citations.

B. Safety Escalation

A deterministic phrase classifier triggers hospital-visible escalation for emergency-like utterances.

C. Standards Alignment

Application-level consent types (HEALTH_RECORD_ACCESS, THIRD_PARTY_MODEL_PROCESSING) are complementary to the FHIR R4 Consent resource [26]; mapping typed grants to FHIR Consent is future work. SMART App Launch v2.2 scopes [27] authorize FHIR API clients; the MCP gateway sits above SMART-style scopes as a tool-mediation and audit boundary for agent invocations, not as a replacement for OAuth/SMART.

IV. Data Collection

We construct a synthetic corpus under examples/evaluation-data/ (Fig. 3). Table I summarizes resource counts. Curated gold patients ($n=5$) expose explicit facts; Synthesia exports ($n=10$, seed 1784242671144) are slimmed to supported FHIR types. Gold QA ($n=25$) includes unanswerable LDL probes. Escalation includes exact ($n=22$) and paraphrase ($n=22$) sets; consent scenarios ($n=8$).

TABLE II
Emergency escalation: exact vs paraphrase.

Split	n	Prec.	Rec.	F1
Exact	22	1.00	1.00	1.00
Paraphrase	22	1.00	0.17	0.29
Combined	44	1.00	0.55	0.71

V. Experiments

We report: (1) lexical Recall@3; (2) live OpenAI RAG with answer-string-required audit; (3) consent-gate stub e2e; (4) exact vs paraphrase safety; (5) Go unit/contract tests.

B. Results

Lexical existence oracle and type-aware Recall@3 are both 1.00. Live RAG answerable accuracy is 0.85 (17/20) with citation presence 1.00 and safe refusal 1.00 on 5 LDL probes (mean 1667 ms, p95 2432 ms). Error analysis: allergy misses in answer text (3) and one medication answer that incorrectly included penicillin (an allergy) alongside albuterol. Exact escalation F1 = 1.00; paraphrase F1 = 0.29; combined F1 = 0.71 (Table II). Consent deny/allow/regrant for HEALTH_RECORD_ACCESS passes; third-party model consent remains a known RAG-pipeline gap.

VI. Discussion and Limitations

Separating MCP and audit from the voice runtime creates a single policy enforcement point. Relative to Hou et al. [22], the gateway mitigates unauthorized tool invocation, weak accountability, and unmediated PHI access, but does not yet address installer spoofing or cryptographic tool-manifest signing. Relative to Polaris [6], we trade proprietary constellation training for an auditable open architecture. Limitations include modest live-RAG gold size, paraphrase brittleness of phrase safety (motivating LLM/embedding classifiers [24], [25]), incomplete third-party consent enforcement, lack of clinician-rated groundedness, and absence of MedAgentBench-style interactive evaluation [10]. Synthetic FHIR under-represents messy clinical narratives. Aligning typed consents with FHIR Consent [26] and composing MCP mediation with SMART scopes [27] are interoperability priorities.

VII. Conclusion

Zorba Health provides an open, consent-aware architecture for voice healthcare agents with FHIR-grounded RAG and MCP-mediated tools, plus a reproducible synthetic evaluation package with honest safety stress results. Future work includes paraphrase-robust escalation, third-party consent enforcement, FHIR Consent / SMART alignment, MedAgentBench and HealthBench-style evaluation, Synthesia-scale gold labels, and clinician review.

Data Availability

Evaluation artifacts and scripts are included in the open-source repository under `examples/evaluation-data/` and `scripts/evaluation/`. Synthea seed: 1784242671144. Draft freeze commit: 2d7429d0.

Conflict of Interest

The author maintains the Zorba Health open-source project and declares no competing financial interests.

Ethics

Evaluation uses synthetic data only. No human subjects or production PHI were used. Outputs are not clinical advice. Framing follows WHO guidance on large multi-modal models for health [2].

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